

NIEL ABHISHEK J DAVID

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PROFESSIONAL SUMMARY

Aspiring Computer Science Engineer with experience building backend applications using Java, Spring Boot, Python, FastAPI, and SQL. Skilled in designing REST APIs, cloud-native applications, and scalable software systems with hands-on experience in AWS, Docker, CI/CD, and database design. Passionate about backend engineering, distributed systems, and building reliable software.

EXPERIENCE

❖ Intern | Indian Institute of Science (IISc), Bengaluru

Feb 2025 – May 2025

- Architected AI-driven microservices using **FastAPI**, **React.js**, and **WebSockets** for MagicMirror² project
- Integrated **Firebase** for user **authentication**, **Storage** and **Firestore**
- Integrated **Gemini LLM** and **Face-Recognition** for personalized UX.
- Followed **Agile methodology** with spring-based progress and cross-functional team and collaboration.

❖ Data Analyst (GAT) | Volvo Group India Pvt Ltd

March 2026 – Present

- Engineered Python (Pandas/NumPy) and SQL data wrangling scripts on Azure Databricks to automate data cleansing and preprocessing for aftermarket planning pipelines, reducing processing times.
- Designed automated ETL workflows using Power Automate and Azure services to streamline cross-system data flows between Dataverse and enterprise storage.
- Developed interactive Power BI dashboards and automated reports connected to SQL databases and Dataverse/SharePoint, transforming raw metrics into key business KPIs.

PROJECTS

❖ [FitTrack Pro](#)

Tech Stack: Spring Boot 3, Spring Security, Postgre SQL, JWT , BCrypt Password Encoder, Docker ,React.js

- Developed an enterprise-grade RESTful backend using Spring Boot, Spring Security, JWT, and PostgreSQL to build secure role-based access control (RBAC), workout scheduling, and reporting APIs using Hibernate/JPA and Flyway
- Enhanced API performance and developer experience using Redis caching and Swagger/OpenAPI documentation within a clean layered architecture applying SOLID principles.
- Built a production-ready system utilizing Docker, Docker Compose, JUnit 5, Mockito, and GitHub Actions CI/CD for automated testing and scalable deployment.

❖ [Digital Wardrobe](#)

Tech Stack: FastAPI, React.js, WebSockets, Gemini LLM, OpenCV, Face-Recognition, Firebase(Database)

- Architected AI-driven microservices using FastAPI and WebSockets to deliver low-latency communication between smart mirror services.
- Implemented secure facial recognition, Firebase Authentication, and cloud-based wardrobe management using Firestore and Firebase Storage.
- Containerized backend services with Docker and integrated Gemini LLM for personalized outfit recommendations and voice-assisted interactions.
- Designed scalable backend services using modular architecture and REST APIs to support real-time communication across distributed components.

❖ Fashion Recommendation System

Tech Stack: TensorFlow, EfficientNetB3, Flask, SQLAlchemy, MLflow, React

- Developed an EfficientNetB3-based deep learning model achieving 92% accuracy for personalized fashion recommendations.
- Designed PostgreSQL schemas with SQLAlchemy for efficient storage of user profiles, measurements, and recommendation history.
- Containerized ML inference services using Docker and automated model validation and deployment workflows with GitHub Actions.

❖ Cloud-Based Attendance System

Tech Stack: React.js, Node.js, Firebase Auth/Cloud firestore, CI/CD, GitHub Actions

- Built a responsive school attendance system with secure login, real-time updates via Firebase Firestore, and student profile management.
- Implemented a modular backend with RBAC authentication and analytics dashboard to visualize attendance patterns.
- Cloud-native solution automating attendance tracking with secure access and real-time updates.

❖ Network Security Firewall Solution

Tech Stack: Python, Tornado, Flask, Distributed Caching, Intrusion Detection Systems

- Developed a rule-based intrusion detection system using Python, Flask, and Tornado to support concurrent traffic inspection and centralized security rule management.
- Built RESTful management APIs allowing administrators to dynamically configure security policies and monitor suspicious network activity.
- Optimized session management using Redis caching and containerized the entire application with Docker for portable, high-throughput deployment.

EDUCATION

- ❖ **B.Tech in Computer Science Engineering**
University, RR Campus, Bengaluru

Jan 2023 – Present PES
CGPA – 7.73

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C#, JavaScript, Bash
- **Frontend Development:** HTML, CSS, React.js
- **Backend Development:** Servlets, Spring IoC & MVC, Spring Boot, Spring Security, Hibernate, JPA, FastAPI, Django, REST APIs, JWT Authentication, ASP.NET Core
- **Database:** MySQL, MongoDB, Oracle SQL(PL/SQL)
- **Machine Learning Libraries:** Pandas, NumPy, TensorFlow, Scikit-Learn
- **Cloud: Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, Docker Compose, GitHub Actions, CI/CD

CERTIFICATIONS

- Beginner's guide to Linux Kernel Development. ([Certificate](#))
- AWS educate Introduction to Cloud. ([Storage](#), [Compute](#), [Serverless](#), [Networking](#))
- Jira Work Management Fundamentals Badge. ([Atlassian University](#))
- JSpiders Java Full Stack + GenAI Course

ACHIEVEMENTS

- Silver Medalist – IndiaSkills Regional 2021 (Industrial Control)
- 5th Place – IndiaSkills National 2022 (Industrial Control)
- Gold Medalist – Karnataka Skills State 2021 (Industrial Control)